



Université
Gustave Eiffel



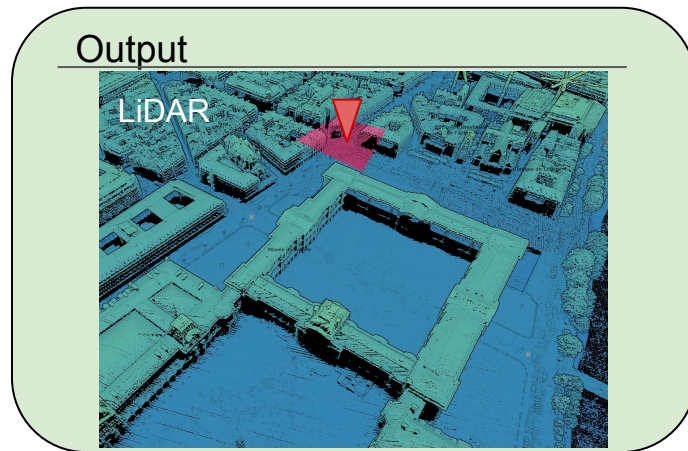
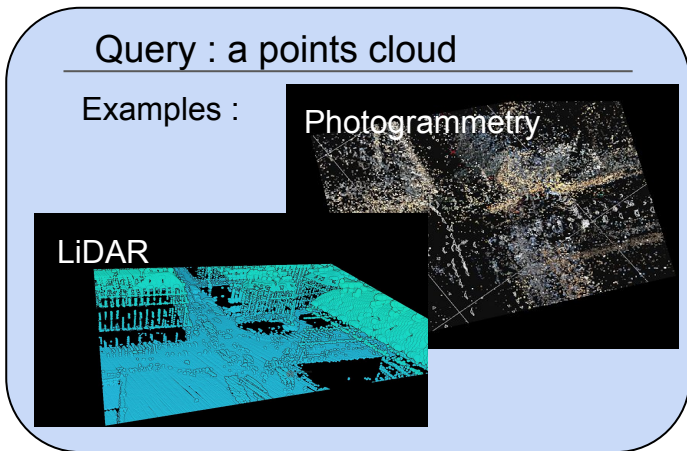
Fast 3D point clouds retrieval for Large-scale 3D Place Recognition

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Context

Objectives

- Fast 3D point clouds retrieval for large-scale place recognition
- Geolocalize a points cloud in a reference dataset
- Adaptable to different types of acquisitions (Photogrammetry, SLAM, LiDAR ...)
- Scalable solution



I. State of the art

LiDAR-based Place Recognition

Inputs:

- 3D data query
- 3D reference dataset with seen scenes

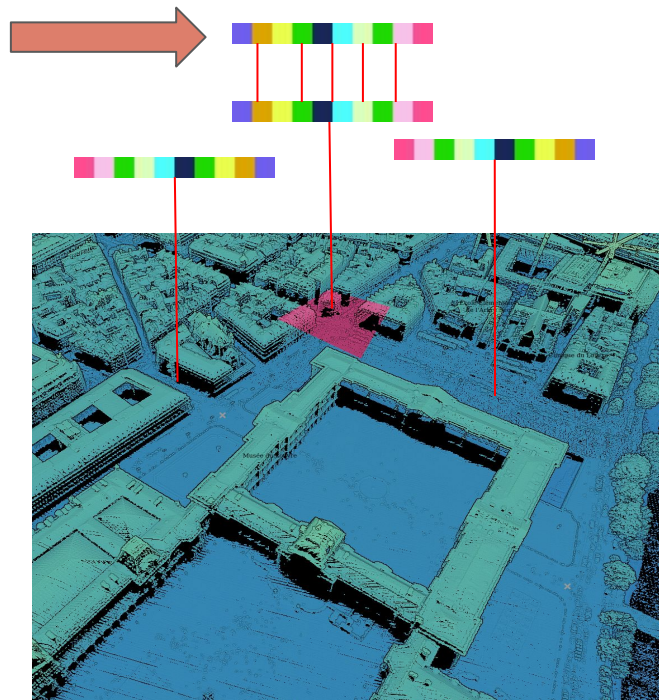
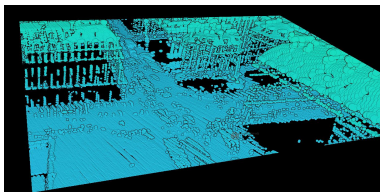
Place recognition in two steps:

Offline

- Reference dataset description (with points cloud descriptors)

Online

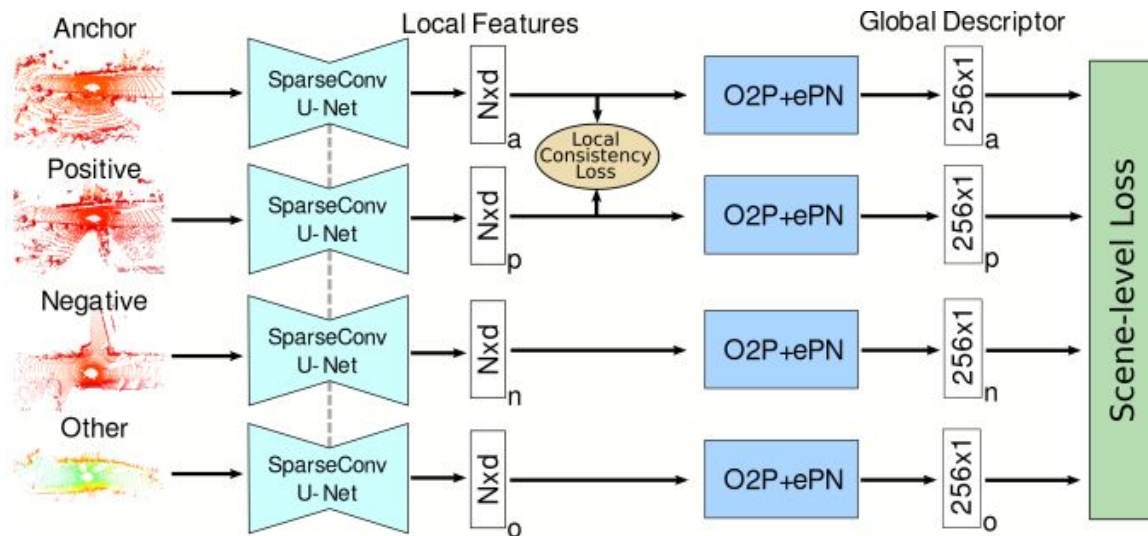
- Query description
- Place retrieval by searching for the nearest descriptors of the query one in the reference dataset



I. State of the art

Example of 3D place recognition approach:

Logg3D-net (Vidanapathirana et al, 2022)



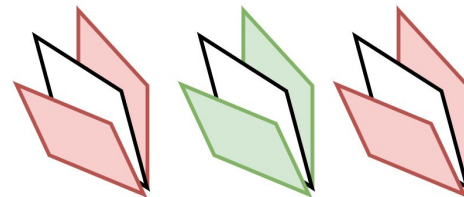
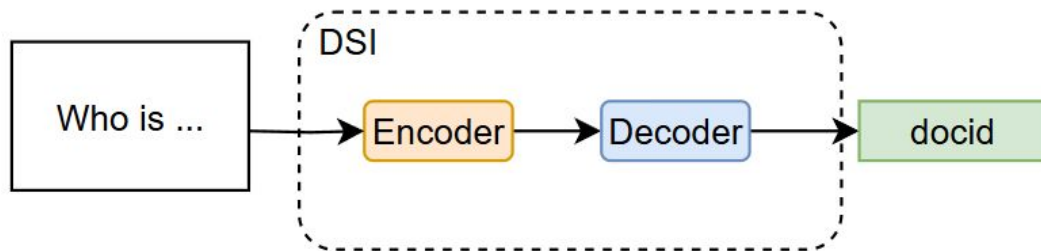
Contributions :

- Local consistency loss
- Second-order pooling for local descriptors aggregation

II. Differentiable Search Index (DSI)

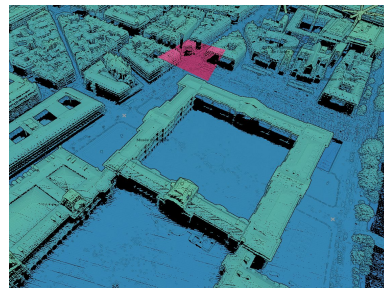
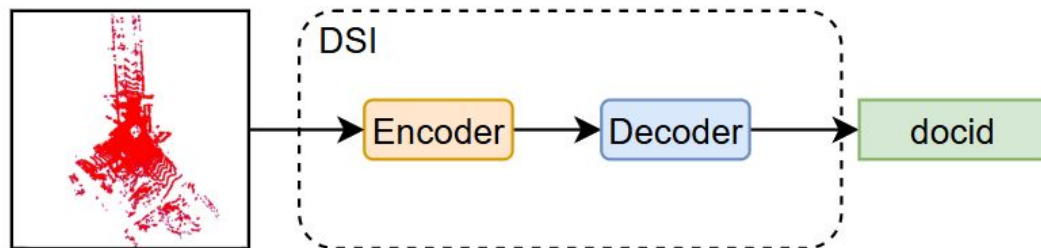
Idea

DSI (Tay et al, 2022) principle on text ...



Return a relevant document ID

... Applied to 3D, **DSI-3D** !

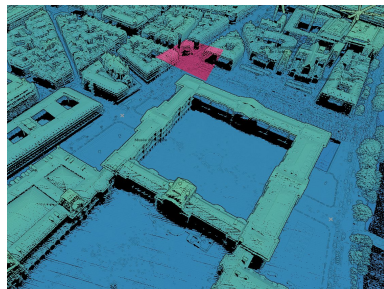
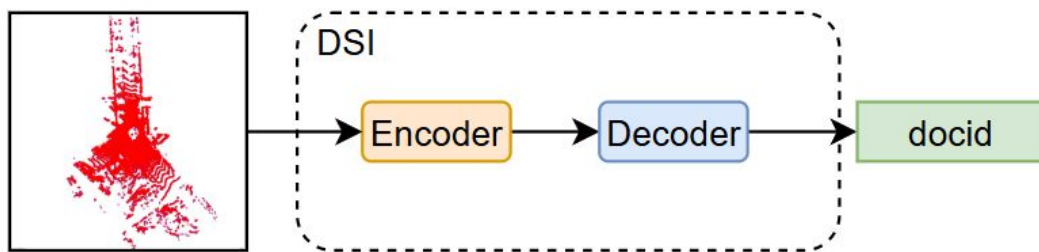


Return a relevant place ID

II. Differentiable Search Index (DSI)

Idea

... Applied to 3D, **DSI-3D** !



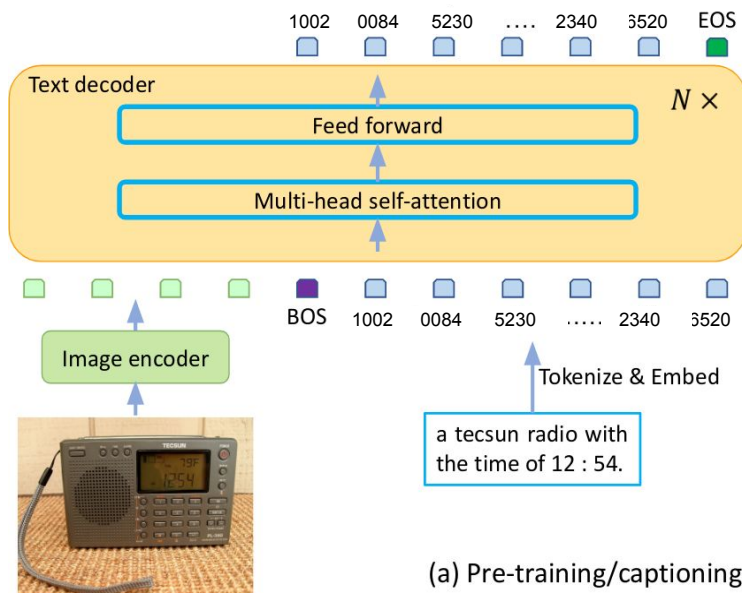
Return a relevant place ID

Main benefits of DSI-3D:

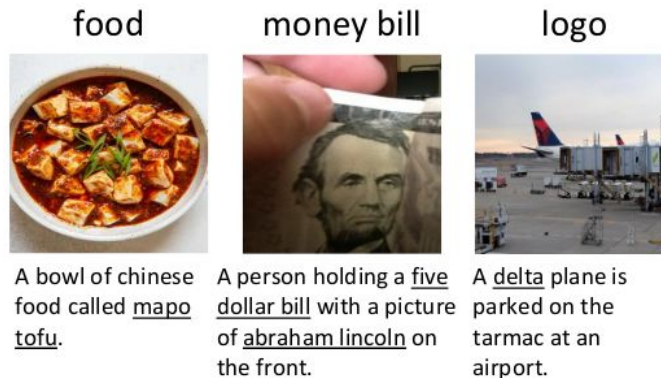
- Retrieval at constant inference time
- Use of a single transformer for the retrieve-then-rank pipeline

III. Vision to text

GIT (Generative Image-to-text Transformer)



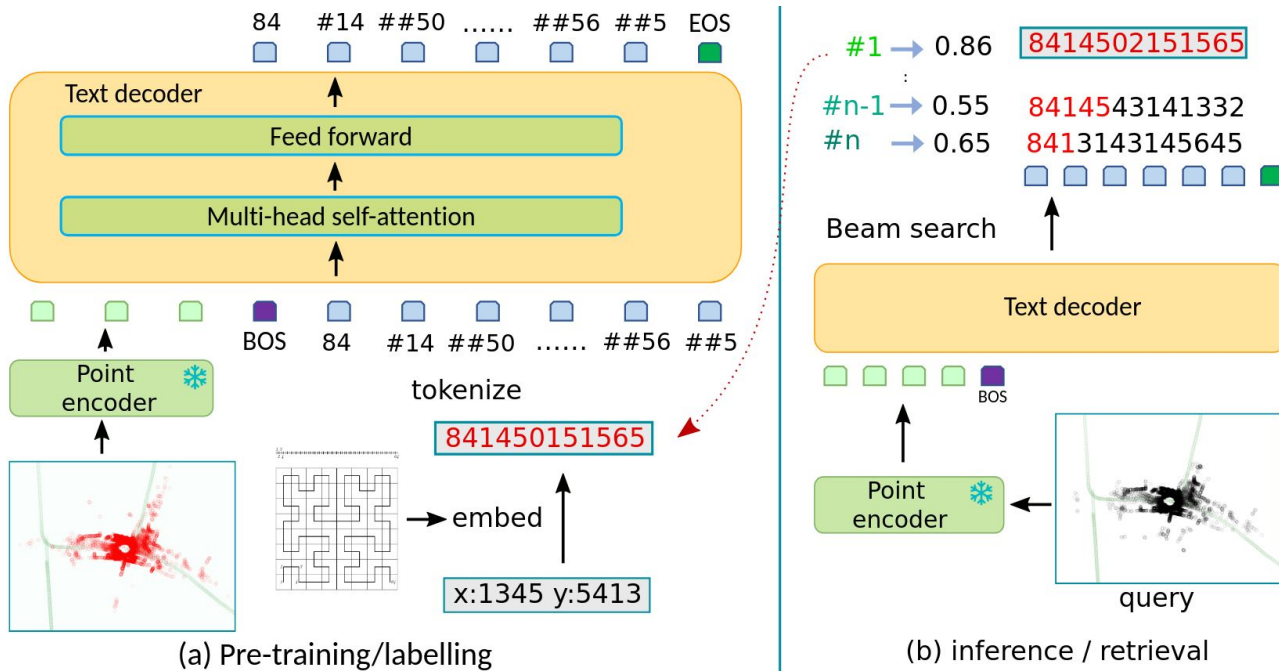
Example of descriptions from an image input



Wang et al, 2023

III. LiDAR to text

GIT (Generative Image-to-text Transformer) adapted to 3D input

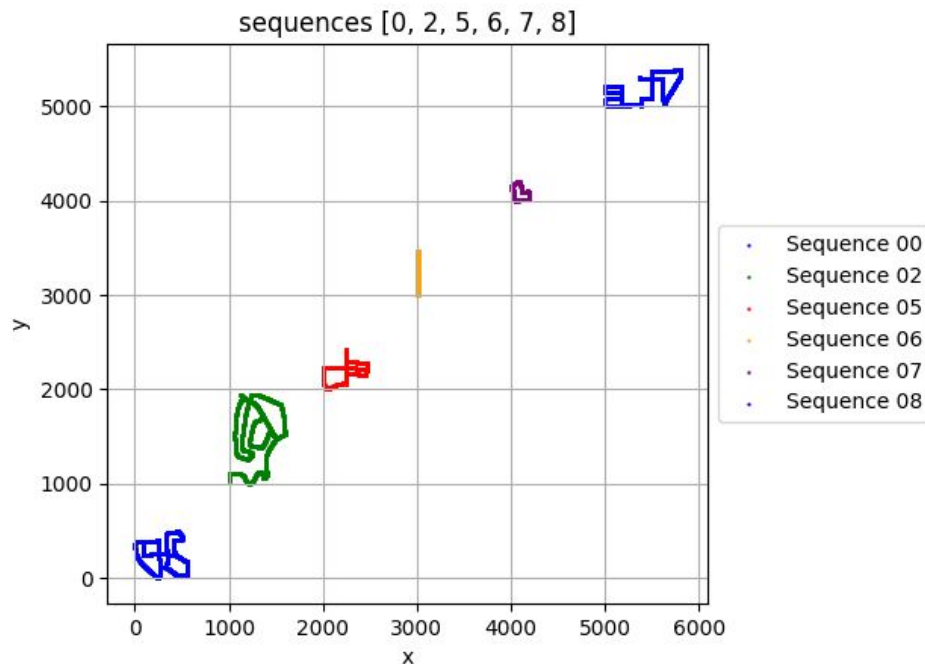


- Labeling step: the text decoder learns to generate the label according to the encoded point cloud.
- Retrieval: a beam search is performed by the text decoder to provide the n most probable solutions.

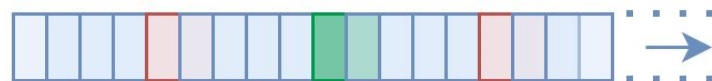
tokens : ■ Point ■ Label ■ Beginning-of-sequence ■ End-of-sequence Networks ❄ Frozen networks

IV. Dataset

KITTI dataset (Geiger et al, 2013)



- Point clouds are acquired sequentially.
- The 6 KITTI sequences with revisits are used (00, 02, 05, 06, 07, 08) and a combination of those 6 sequences named 22.
- The dataset is divided as follow :



 train set (80%)

 val set (10%)

 eval set (10%)

V. Indexing: Naively Structured String Identifiers

Indexing strategies:

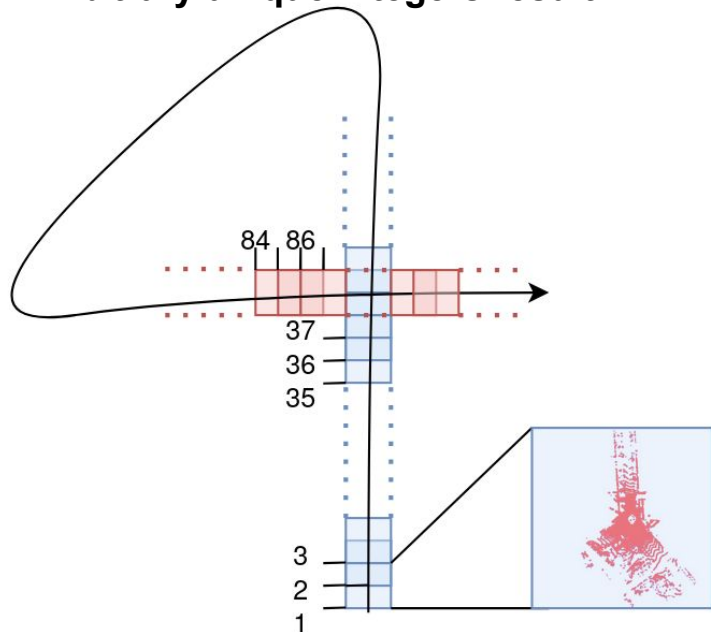
As DSI is based on a text-to-text transfer transformer, it decodes generated tokens into text. This raises the question of **what is the best text representation of locality**.

We explored the following strategies:

- Naively Structured String Identifiers
- Semantically Structured Identifiers
- Positional Structured identifiers
- Positional Structured identifiers (Hilbert curve)

V. Indexing: Naively Structured String Identifiers

Arbitrary unique integers result in:



Advantages:

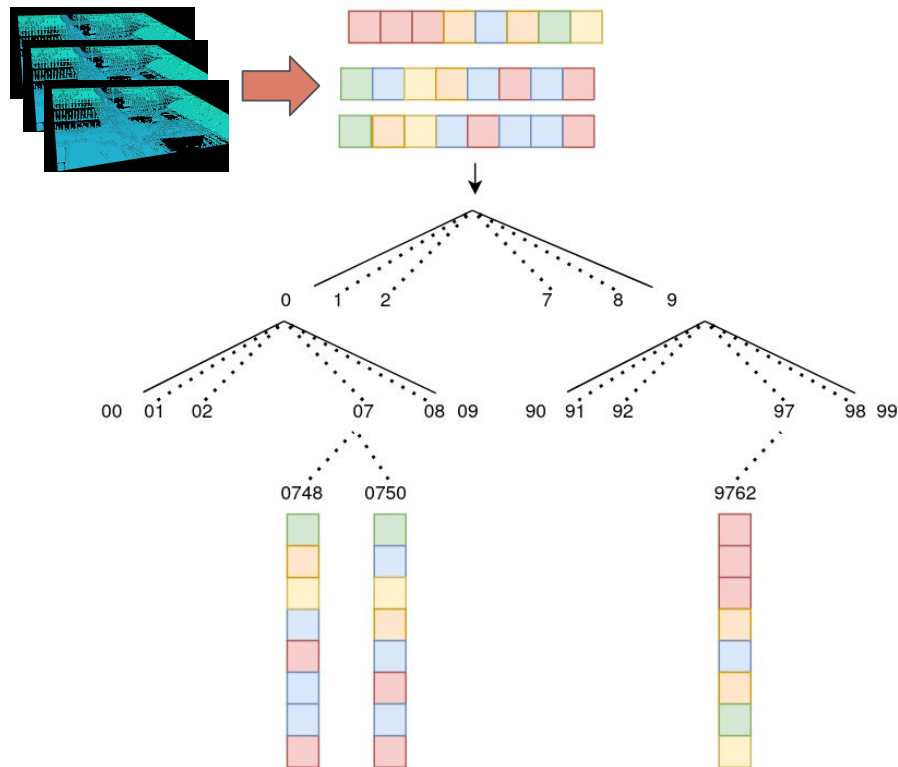
- Easy to implement
- “time” proximity

Limits:

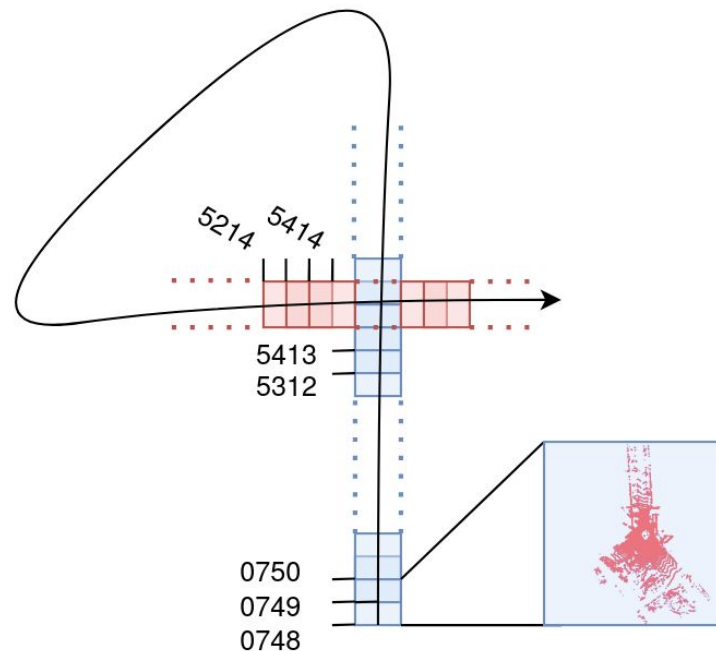
- Revisits of a scene may have very different indexes

V. Indexing: Semantically Structured Identifiers

Hierarchical clustering of 3D descriptors

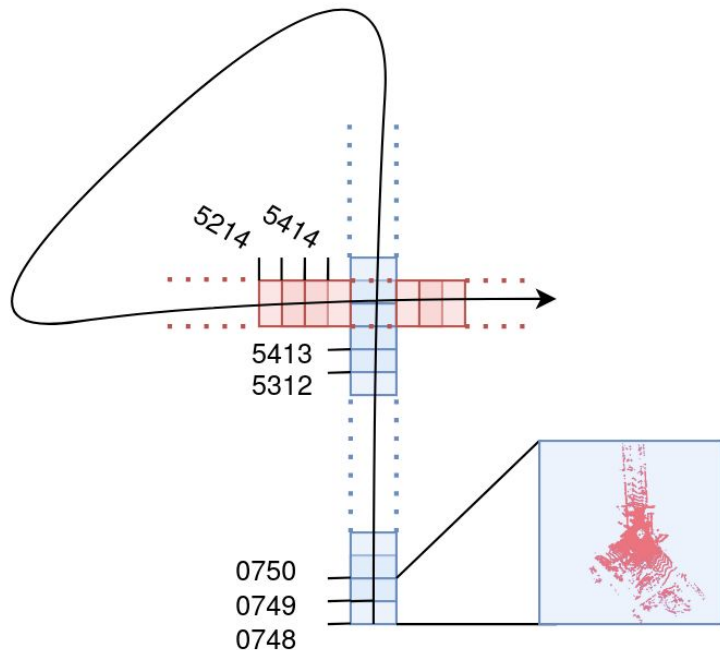


Result in:



V. Indexing: Semantically Structured Identifiers

Hierarchical clustering of 3D descriptors



Advantages:

- Compact indexes
- Contains descriptor information.

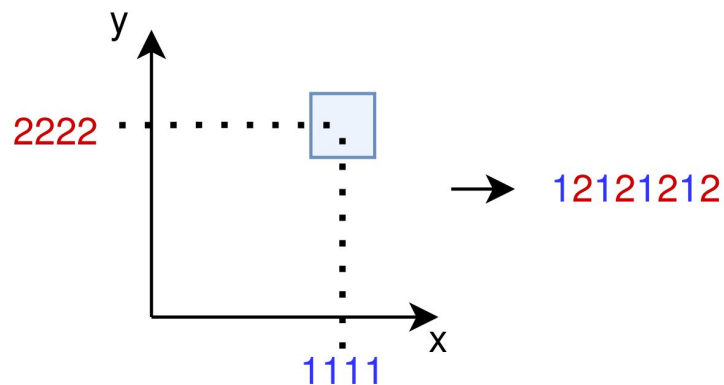
Limits:

- No geographical informations

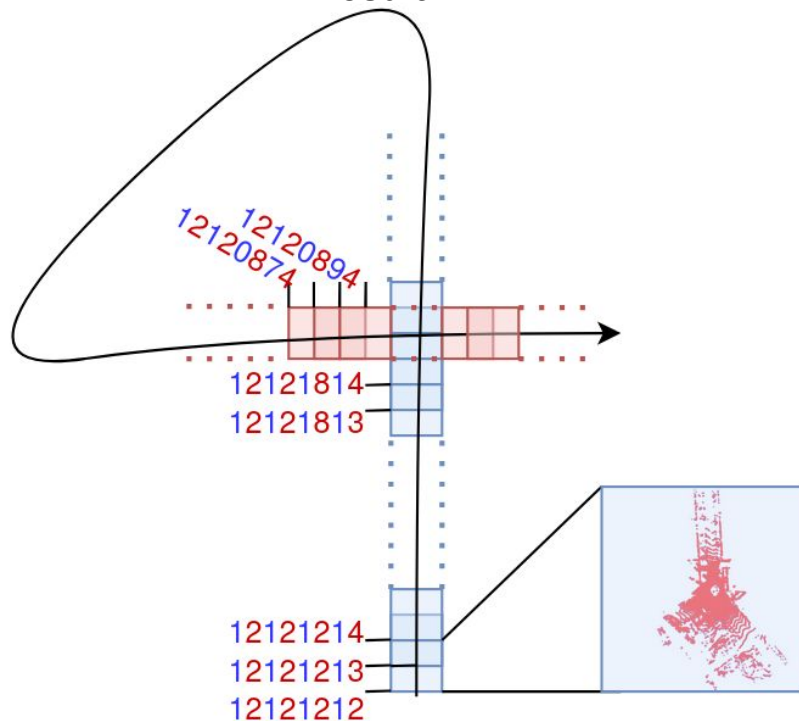
V. Indexing: Positional Structured identifiers with GPS

Tailor-made for georeferenced data:

- Center coordinates of each point cloud are used to create a unique index by interlacing their digit string.

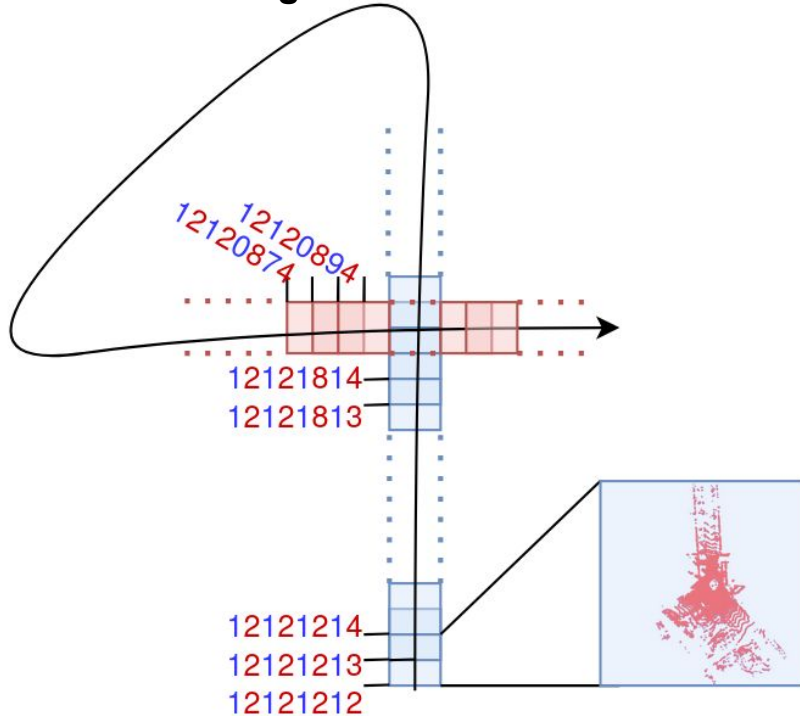


Result in:



V. Indexing: Positional Structured identifiers with GPS

Tailor-made for georeferenced data:



Advantages:

- Contains location information.

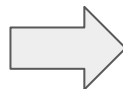
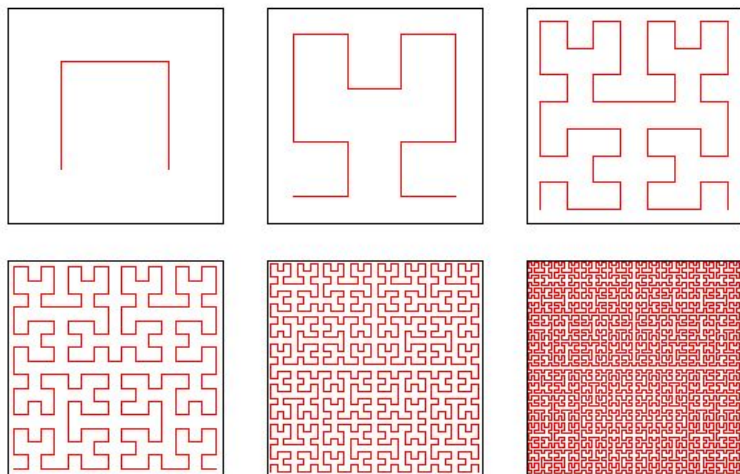
Limits:

- A direction (here X) has more weight than the other (Y).
- Longer indexes

V. Indexing: Positional Structured identifiers (Hilbert curve)

Tailor-made for georeferenced data:

- Groups of fractals function that map a N dimensional space to a 1D distance.



Result in:

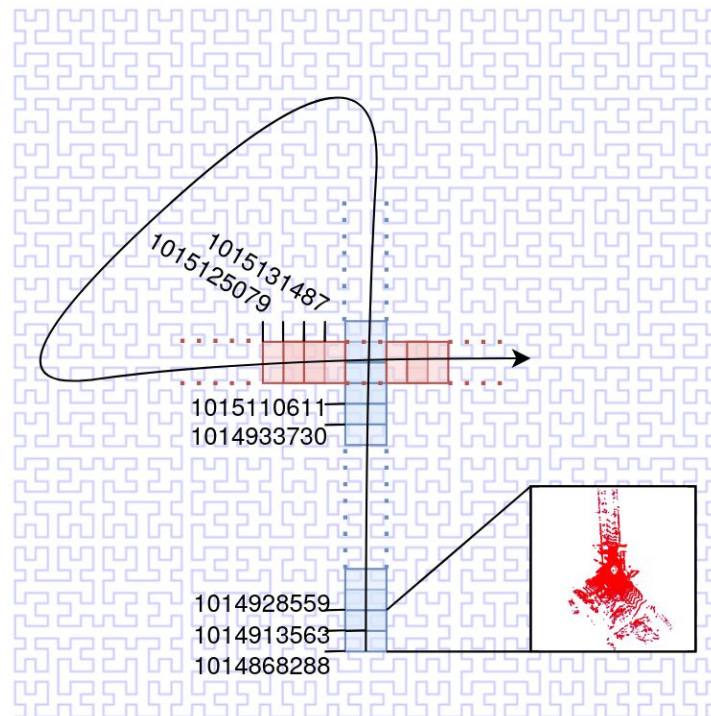
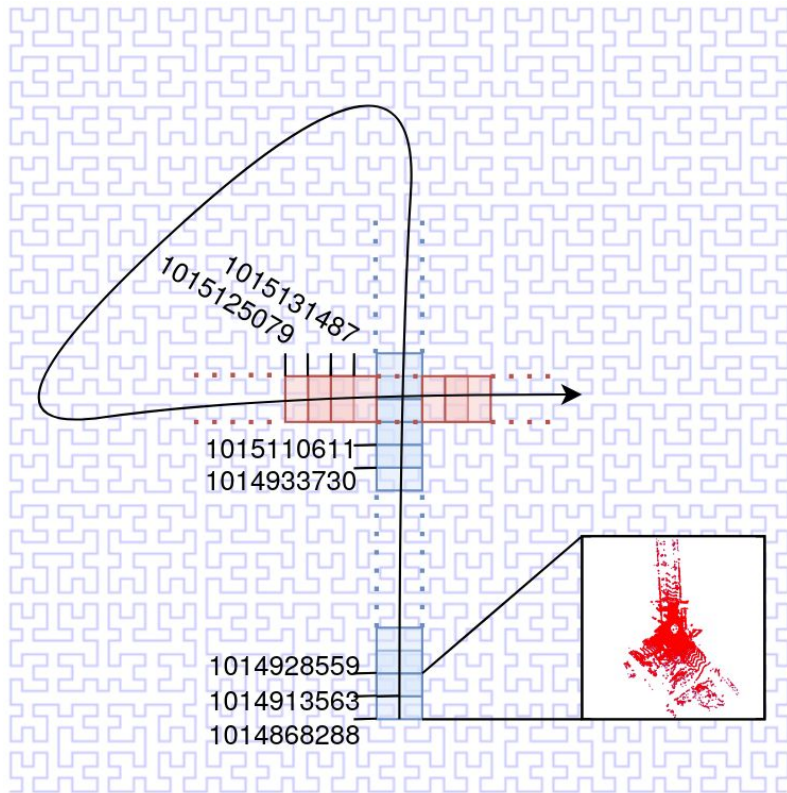


Figure from : https://fr.wikipedia.org/wiki/Courbe_de_Hilbert

V. Indexing: Positional Structured identifiers (Hilbert curve)



Advantages:

- Contains location information.
- Best conservations of locality (Faloutsos et al, 1989)

Limits:

- Longer indexes.

VI. Evaluation of the indexing strategies

Hits@1 scores for DSI-3D

A revisit is considered correct if it is within 3 meters of the ground truth.

model	00	02	05	06	07	08
ours (label)	0.8132	0.7645	0.8831	0.6306	0.7928	0.8235
ours (Hierar)	0.9319	0.8630	0.9387	0.9279	0.9640	0.9387
ours (GPS)	0.9626	0.9507	0.9783	0.9369	0.9909	0.9705
ours (Hilbert)	0.9846	0.9550	0.9819	0.9640	0.9910	0.9828

The Hilbert Curve indexing strategy is the most efficient at retrieving neighbours.

VI. Evaluation of the indexing strategies

F1max scores for DSI-3D

A revisit is considered correct if it is within 3 meters of the ground truth and is below the model's confidence thresholds.

model	00	02	05	06	07	08
ours (label)	0.8991	0.8774	0.7906	0.7954	0.8844	0.9044
ours (Hierar)	0.9635	0.9316	0.9684	0.9626	0.9817	0.9684
ours (GPS)	0.9809	0.9758	0.9891	0.9674	0.9955	0.9851
ours (Hilbert)	0.9933	0.9781	0.9909	0.9817	0.9955	0.9913

The Hilbert Curve indexing strategy is also the most performant in terms of F1 score.

VI. Complexity evaluation

There are two search strategies:

Exhaustive search

This brute force approach evaluates all possible candidates, ensuring the best match but with high computational cost.

- LoGG3D-Net using cosine distance between N descriptors of time complexity $\mathbf{O(N)}$

Approximate search

Approximate search trades the certainty of finding the exact best match for improved speed.

- LoGG3D-Net with Locality-sensitive hashing (LHS) (Pan et al, 2024), by computing a binary index for each descriptor, reducing time complexity to $\mathbf{O(N^p)}$, where $\mathbf{0 \leq p \leq 1}$.
- DSI-3D relies on model inference approximate search via beam search, with constant $\mathbf{O(1)}$ complexity due to fixed number of tokens, number of beams and vocabulary size.

VI. Complexity evaluation

Comparison of Hits@1 scores between DSI-3D and approximative LoGG3D-Net with (LHS) (Pan et al, 2024)

A revisit is considered correct if it is within 3 meters of the ground truth.

model	00	02	05	06	07	08	22
LoGG3D-Net	0.9956	0.9722	1.0	0.9910	1.0	0.9852	0.9852
LoGG3D-Net + LHS (256)	0.9890	0.9572	0.9892	0.9550	0.9820	0.9755	0.9693
ours (Hilbert)	0.9846	0.9550	0.9819	0.9640	0.9910	0.9828	0.9644

The Hilbert Curve indexing strategy is slightly better than LoGG3D-Net+LHS by **0.1%**.

VI. Complexity evaluation

Comparison of Hits@1 scores between DSI-3D and approximative LoGG3D-Net with (LHS) (Pan et al, 2024)

A revisit is considered correct if it is within 3 meters of the ground truth and is below the model's confidence thresholds.

model	00	02	05	06	07	08	22
LoGG3D-Net	0.9956	0.9859	1.0	0.9954	1.0	0.9974	0.9925
LoGG3D-Net + LHS (256)	0.9945	0.9792	0.9946	0.9815	0.9626	0.9888	0.9850
ours (Hilbert)	0.9933	0.9781	0.9909	0.9817	0.9955	0.9913	0.9819

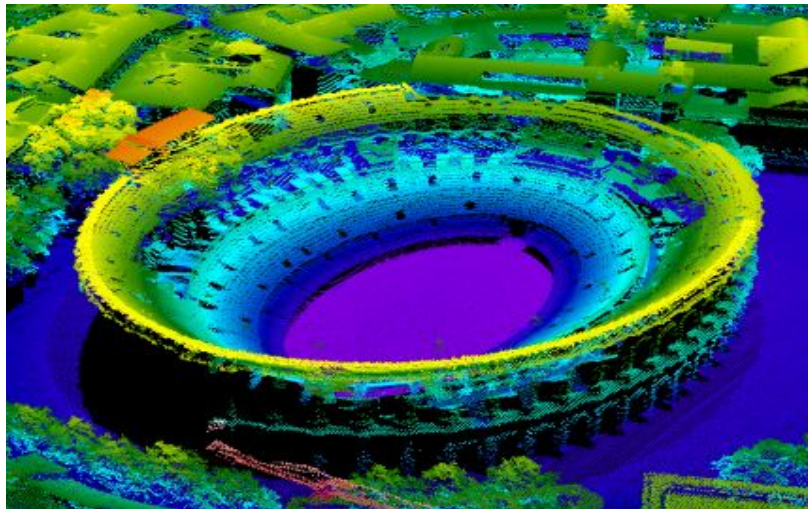
The Hilbert Curve indexing strategy is slightly better than LoGG3D-Net+LHS by **0.4%**.

VII. Conclusion

- While exhaustive search has the highest score in terms of retrieval quality, it is not competitive for large-scale retrieval as its time complexity $O(N)$ scales with the dataset size.
- Approximate search with LoGG3D-Net+LSH offers a trade-off between quality and speed with a time complexity $O(N^p)$, where $0 \leq p \leq 1$ but also becomes slower for larger dataset.
- DSI-3D has retrieval time of **0.09s with a retrieval complexity in $O(1)$**
 - In its current implementation, it becomes faster than exhaustive search when the database contains **136k** scenes. With the same density as KITTI, this covers an area of approximately **11km²**.

VII. Perspectives

- Up scale the training of the model to cover a larger area.
- Conduct experiments on LiDAR HD data from IGN.
- Adapt to multi-source place recognition descriptors.



LiDAR HD: (IGN, 2023)

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